

Student Worksheet

AI-Assisted Critical Review of Your Research Introduction

Course: _____

Student Name: _____

Date: _____

Title of Paper: _____

Purpose of this Activity

This worksheet teaches you how to use AI as a **critical thinking tool**, not a shortcut.

You will:

- audit your introduction like a peer reviewer
- identify weak claims
- test your research gap
- improve scientific reasoning
- revise with intention

AI is acting as a **skeptical academic auditor**.

Your job is to interpret and apply its feedback.

Step 1: Run the AI Audit

Copy your full introduction and paste it into the AI using the instructor prompt.

Then paste the AI output below.

AI Output (Paste here)

[Paste full AI audit response]

Step 2: Structural Understanding

Read the AI's **Structural Map** section.

For each part, summarize what the AI is saying in your own words.

Section	What AI Says Is Weak or Missing	Your Interpretation
Hook + Background		
Problem Definition		
Research Gap		
Aim/Objectives		
Proposed Solution		
Roadmap		

👉 Reflection:

Which section is your weakest? Why?

Step 3: Evidence Discipline

Look at the Evidence & Fact-Check Table.

Answer:

1. Which claims require evidence?

2. Which claims are risky if wrong?

3. Which claims are too broad or exaggerated?

👉 Rewrite ONE weak claim using safer scientific language:

Original:

Rewritten:

Step 4: Research Gap Check

A strong research gap is:

✓ specific

✓ realistic

✓ measurable

✓ defensible

Answer:

Is your gap specific?

☐ Yes ☐ No

If no, rewrite it:

Does your gap claim total novelty?

☐ Yes ☐ No

Rewrite safely:

Step 5: Scientific Reasoning Check

Answer YES or NO:

Does your introduction:

☐ confuse correlation with causation

☐ overgeneralize

☐ make hidden assumptions

☐ promise unrealistic outcomes

☐ lack reproducibility clarity

Explain one issue in your own words:

Step 6: Aim & Objectives

Write your revised aim:

Write 3 SMART objectives:

1.

2.

3.

Step 7: Revision Planning

Based on AI feedback:

Top 3 changes you must make:

1.

2.

3.

Structural issues:

Cosmetic issues:

Revision order:

First →

Second →

Third →

Step 8: Reflection

What did the AI reveal about your writing that you did not notice?

What will you do differently in future research writing?

Submission Instructions

Submit:

- ✓ This completed worksheet
- ✓ Your revised introduction
- ✓ Original introduction (before revision)

Key Lesson

AI is not the authority.

AI is a mirror.

Your job is to:

- question claims
- demand evidence
- reduce exaggeration
- clarify logic
- strengthen argument

That is scientific thinking.